Discrete Structures: The Foundations – Logic and Proofs: Part 3

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Outline

- Propositional Logic
- Applications of Propositional Logic
- Propositional Equivalences
- Predicates and Quantifiers
- Nested Quantifiers
- Rules of Inference
- Introduction to Proofs

Introduction 1

- Can you define *exclusive or* by using negation, conjunction and disjunction?
  - $\boldsymbol{p} \oplus \boldsymbol{q}$ can be rewritten as $formula(p, q, \neg, \wedge, \vee)$?
  - p or q, but not both
    - The key would be how to translate the word "but"
    - This feels like to remove one of the true cases
      - We don't have the connectives to "remove" a case
      - Then what ???
    - How about we first translate these two parts
      - Part 1: $p \vee q$
      - Part 2: $\neg(p \wedge q)$
    - Try each connective to find a fit

Introduction 2

- Can you define exclusive or by using negation, conjunction and disjunction?
  - p or q, but not both
    - Try each connective to find a fit
    - How about conjunction: p or q, and not both ?
      - $(p \vee q) \wedge \neg(p \wedge q)$ – How to verify? Truth table !!!

p	q	$p \vee q$	$\neg(p \wedge q)$	$(p \vee q) \wedge \neg(p \wedge q)$	$p \oplus q$
T	T	T	F	F	F
T	F	T	T	T	T
F	T	T	T	T	T
F	F	F	T	F	F

Logical Equivalence 1

- The compound propositions p and q are called ***logically equivalent*** if $p \leftrightarrow q$ is always true. The notation, $p \equiv q$, denotes that p and q are logically equivalent.
- One way to determine whether two compound propositions are equivalent is to use a truth table.
  - Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.

Logical Equivalence 2

- Some common sense

Equivalence	Name
$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity laws
$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

Tautology, Contradiction, and Contingency

- A ***tautology*** is a proposition which is always true.
  - Example: $p \vee \neg p$
- A ***contradiction*** is a proposition which is always false.
  - Example: $p \wedge \neg p$
- A ***contingency*** is a proposition which is neither a tautology nor a contradiction
  - Example: $p \wedge q$
- **Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.**
- **How to evaluate: Truth table!**

Examples of Logical Equivalence 1

p	q	$\neg p$	$\neg p \vee q$	$(\neg p \vee q) \leftrightarrow (p \rightarrow q)$	$p \rightarrow q$
T	T	F	T	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	T	T

Tautology

$\neg p \vee q \equiv p \rightarrow q$ **because** $(\neg p \vee q) \leftrightarrow (p \rightarrow q)$ **is true for every value of** p **and** q**.**

- This is an important conclusion: convert implication to basic conjunctions and disjunctions.

Basic Logical Equivalences with conditional statements
$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

Examples of Logical Equivalence 2

- $p \rightarrow q \equiv \neg p \vee q$
  - If that country doesn't invest in USA, then they will suffer from increased tariff.
  - That country invests in USA, or they will suffer increased tariff.
- $p \rightarrow q \equiv \neg q \rightarrow \neg p$
  - Recall contrapositive
  - If students passes the exam, then they scored at least 60%.
  - If students scored less than 60%, then they did not pass the exam.

Examples of Logical Equivalence 3

- $\boldsymbol{p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)}$
  - Recall definition of biconditional (if and only if)
  - Access happens exactly when the password is correct
  - p → q: If the correct password is entered, access is granted.
    AND
    q → p: If access is granted, then the correct password was entered.
- $\boldsymbol{p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)}$
  - The switch and the light always have the same state.
  - The switch is ON and the light is ON
    OR
    The switch is OFF and the light is OFF

Examples of Logical Equivalence 4

p	q	$(p \wedge q)$	$\neg(p \wedge q)$	$\neg(p \wedge q) \leftrightarrow \neg p \vee \neg q$	$\neg p \vee \neg q$	$\neg p$	$\neg q$
T	T	T	F	T	F	F	F
T	F	F	T	T	T	F	T
F	T	F	T	T	T	T	F
F	F	F	T	T	T	T	T

Tautology

$\neg(p \wedge q) \equiv \neg p \vee \neg q$ **because** $\neg(p \wedge q) \leftrightarrow \neg p \vee \neg q$ **is true for every value of** p **and** q**.**

- This is the proof for the first one in De Morgan's Laws.

De Morgan's Laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$
$\neg(p_1 \vee p_2 \vee \cdots \vee p_n) \equiv (\neg p_1 \wedge \neg p_2 \wedge \cdots \wedge \neg p_n)$
$\neg(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \equiv (\neg p_1 \vee \neg p_2 \vee \cdots \vee \neg p_n)$

- Just flip every part: flip between p and ¬p, ∨ and ∧.

Examples of Logical Equivalence 5

- $\neg(\boldsymbol{p_1} \vee \boldsymbol{p_2} \vee \cdots \vee \boldsymbol{p_n}) \equiv (\neg\boldsymbol{p_1} \wedge \neg\boldsymbol{p_2} \wedge \cdots \wedge \neg\boldsymbol{p_n})$
  - $\boldsymbol{p_1}$: "The student passed Chemistry"
    $\boldsymbol{p_2}$: "The student passed Physics"
    $\boldsymbol{p_3}$: "The student passed Math"
  - "The student didn't pass any of the subjects."
    $$\neg(\boldsymbol{p_1} \vee \boldsymbol{p_2} \vee \boldsymbol{p_3})$$
  - "The student failed every subject."
    $$(\neg\boldsymbol{p_1} \wedge \neg\boldsymbol{p_2} \wedge \neg\boldsymbol{p_3})$$
- $\neg(\boldsymbol{p_1} \wedge \boldsymbol{p_2} \wedge \cdots \wedge \boldsymbol{p_n}) \equiv (\neg\boldsymbol{p_1} \vee \neg\boldsymbol{p_2} \vee \cdots \vee \neg\boldsymbol{p_n})$
  - "The student didn't pass all of the subjects."
    $$\neg(\boldsymbol{p_1} \wedge \boldsymbol{p_2} \wedge \boldsymbol{p_3})$$
  - "The student failed Chemistry or Physics or Math."
    $$(\neg\boldsymbol{p_1} \vee \neg\boldsymbol{p_2} \vee \neg\boldsymbol{p_3})$$

Practices 1

- 1. Use De Morgan's laws to find the negation of following statements
  - Jan is rich and happy.
  - Carlos will bicycle or run tomorrow.
  - Mei walks or takes the bus to class.
  - Ibrahim is smart and hard working.

Practices 2

- 1. Use De Morgan's laws to find the negation of following statements

  - Jan is rich and happy.
    - Jan is not rich, or Jan is not happy.
  - Carlos will bicycle or run tomorrow.
    - Carlos will not bicycle, and will not run tomorrow.
  - Mei walks or takes the bus to class.
    - Mei does not walk to class, and does not take the bus to class.
  - Ibrahim is smart and hard working.
    - Ibrahim is not smart, or is not hard working.

Practices 3

- 2. For each of these compound propositions, find an equivalent compound proposition that does not involve conditionals.
  - $p \rightarrow \neg q$
  - $(p \rightarrow q) \rightarrow r$
  - $(\neg q \rightarrow p) \rightarrow (p \rightarrow \neg q)$

Practices 4

- 2. For each of these compound propositions, find an equivalent compound proposition that does not involve conditionals.

  - $p \rightarrow \neg q$
    - $\neg p \vee \neg q$
  - $(p \rightarrow q) \rightarrow r$
    - $(p \wedge \neg q) \vee r$
  - $(\neg q \rightarrow p) \rightarrow (p \rightarrow \neg q)$
    - $(\neg q \wedge \neg p) \vee (\neg p \vee \neg q)$
    - Optionally: the second part already covers all cases of the first, so this simplifies to $\neg p \vee \neg q$
      - Think them as columns in a truth table

Extension 1

- **1. Backward thinking:**
  - Is $(p \vee q) \wedge (\neg p \vee r) \rightarrow (p \vee r)$ a contingency, a contradiction, a tautology or an implication?
    - Can you quickly make a call, without generating the entire truth table?

Extension 2

- Is $(p \vee q) \wedge (\neg p \vee r) \rightarrow (p \vee r)$ a contingency, a contradiction, a tautology or an implication?
  - **Solution**: Always true/false? Or not always? – Find examples.
    - Based on the precedence, "→" will be lastly parsed.
    - Based on the definition of implication, if the left hand side $(p \vee q) \wedge (\neg p \vee r)$ is false, then the entire expression would be true. This needs $p \vee q$ or $\neg p \vee r$ is false. We can easily make p and q false to get it. **So we have a true case**.
    - When $(p \vee q) \wedge (\neg p \vee r)$ is true and $p \vee r$ is false, the entire expression would be false. To make $p \vee r$ false, we need both p and r false, and thus we need q to be true to make $(p \vee q) \wedge (\neg p \vee r)$ true. So we have a false case.
    - $(p \vee q) \wedge (\neg p \vee r) \rightarrow (p \vee r)$ can be true or false – contingency.
    - We just need two cases, instead of all eight cases.

Extension 3

- **2. Some other advanced logical equivalences**

Other Logical Equivalences	
$(\mathbf{p} \vee q) \vee \mathbf{r} \equiv \mathbf{p} \vee (\mathbf{q} \vee \mathbf{r})$ $(\mathbf{p} \wedge q) \wedge \mathbf{r} \equiv \mathbf{p} \wedge (\mathbf{q} \wedge \mathbf{r})$	**Associative laws**
$\mathbf{p} \vee (\mathbf{q} \wedge r) \equiv (\mathbf{p} \vee q) \wedge (\mathbf{p} \vee \mathbf{r})$ $\mathbf{p} \wedge (\mathbf{q} \vee r) \equiv (\mathbf{p} \wedge q) \vee (\mathbf{p} \wedge \mathbf{r})$	**Distributive laws**
$\neg(\mathbf{p} \wedge \mathbf{q}) \equiv \neg\mathbf{p} \vee \neg\mathbf{q}$ $\neg(\mathbf{p} \vee \mathbf{q}) \equiv \neg\mathbf{p} \wedge \neg\mathbf{q}$	**De Morgan's laws**
$\mathbf{p} \vee (\mathbf{p} \wedge q) \equiv \mathbf{p}$ $\mathbf{p} \wedge (\mathbf{p} \vee \mathbf{q}) \equiv \mathbf{p}$	**Absorption laws**
$\mathbf{p} \vee \neg\mathbf{p} \equiv \mathbf{T}$ $\mathbf{p} \wedge \neg\mathbf{p} \equiv \mathbf{F}$	**Negation laws**

Extension 4

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ **because** $p \vee (q \wedge r) \leftrightarrow (p \vee q) \wedge (p \vee r)$ **is true for every value of** p**,** q **and** r**.**

Distributive laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$
$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

Extension 5

- **3. Use advanced equivalences, without the fundamental truth table**
  - Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Expression	Justification
$(p \wedge q) \rightarrow (p \vee q)$	Original expression
$\neg(p \wedge q) \vee (p \vee q)$	Table 3 $(\boldsymbol{p} \rightarrow \boldsymbol{q}) \equiv (\neg \boldsymbol{p} \vee \boldsymbol{q})$
$(\neg p \vee \neg q) \vee (p \vee q)$	De Morgan's law $\neg(\boldsymbol{p} \wedge \boldsymbol{q}) \equiv \neg \boldsymbol{p} \vee \neg \boldsymbol{q}$
$\neg p \vee \neg q \vee p \vee q$	Restatement removing parenthesis
$\neg p \vee p \vee \neg q \vee q$	Commutative law $p \vee q \equiv q \vee p$
$(\neg p \vee p) \vee (\neg q \vee q)$	Associative law $(p \vee q) \vee r \equiv p \vee (q \vee r)$
$(p \vee \neg p) \vee (q \vee \neg q)$	Commutative law $p \vee q \equiv q \vee p$ applied twice
$\mathbf{T} \vee \mathbf{T}$	Negation law $p \vee \neg p \equiv \mathbf{T}$ applied twice.
$\mathbf{T}$	Domination law $p \vee \mathbf{T} \equiv \mathbf{T}$